AI ENGINEERING · 2026

# FROM PROMPT TO GRAPH

*five layers we've wrapped around the model, and why each one had to exist*

# What we're covering

| | | |
|---|---|---|
| 01 | **PROMPT** | *when the prompt was the whole program* |
| 02 | **CONTEXT** | *what does the model actually get to see?* |
| 03 | **HARNESS** | *what surrounds the model?* |
| 04 | **LOOP** | *what happens after the first answer?* |
| 05 | **GRAPH** | *or is a graph just a loop that branches?* |

*Each one wraps the last. None of them replaced it.*

01

# PROMPT

*when the prompt was the whole program*

2020

# NO GRADIENTS. JUST EXAMPLES.

*GPT-3 — "Language Models are Few-Shot Learners." The model learns from what's in the prompt, and nothing is trained.*

# 3.2×

Chain-of-thought prompting tripled PaLM 540B's accuracy on GSM8K.

*17.9% → 56.9%, from wording alone. Wei et al., Google Research, 2022.*

# Training language models to follow instructions with human feedback

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OpenAI

# A 100× SMALLER MODEL WON.

*InstructGPT: labelers preferred the 1.3B tuned model over the 175B base. Following intent beat raw scale.*

deepseek

# DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

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## Abstract

General reasoning represents a long-standing and formidable challenge in artificial intelligence. Recent breakthroughs, exemplified by large language models (LLMs) (Brown et al., 2020; OpenAI, 2023) and chain-of-thought prompting (Wei et al., 2022b), have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent upon extensive human-annotated demonstrations, and models' capabilities are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labeled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification, and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions, and STEM fields, surpassing its counterparts trained via conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically harnessed to guide and enhance the reasoning capabilities of smaller models.

# 74.4%

Trained to reason, o1 scored 74.4% on AIME 2024. GPT-4o, prompted, scored 9.3%.

*pass@1 at maximal test-time compute. OpenAI, September 2024.*

# STOP TELLING IT TO THINK.

*OpenAI now advises against chain-of-thought prompts. The 2022 trick became a training objective.*

# But it didn't die. It moved.

STILL BRITTLE

**Reword the question, move the score 12%.**

*One perturbation flips model rankings in 63% of cases. Brittlebench, 2026.*

STILL ADVISED

**Examples are still "strongly advised."**

*Anthropic on context engineering, September 2025.*

*What got absorbed was hand-written chain-of-thought. The rest changed name.*

what a prompt still can't be

# static. blind. alone.

*However well it reads you, it knows nothing you didn't tell it — and there is a hard limit on how much you can tell it.*

02

# CONTEXT

*what does the model actually get to see?*

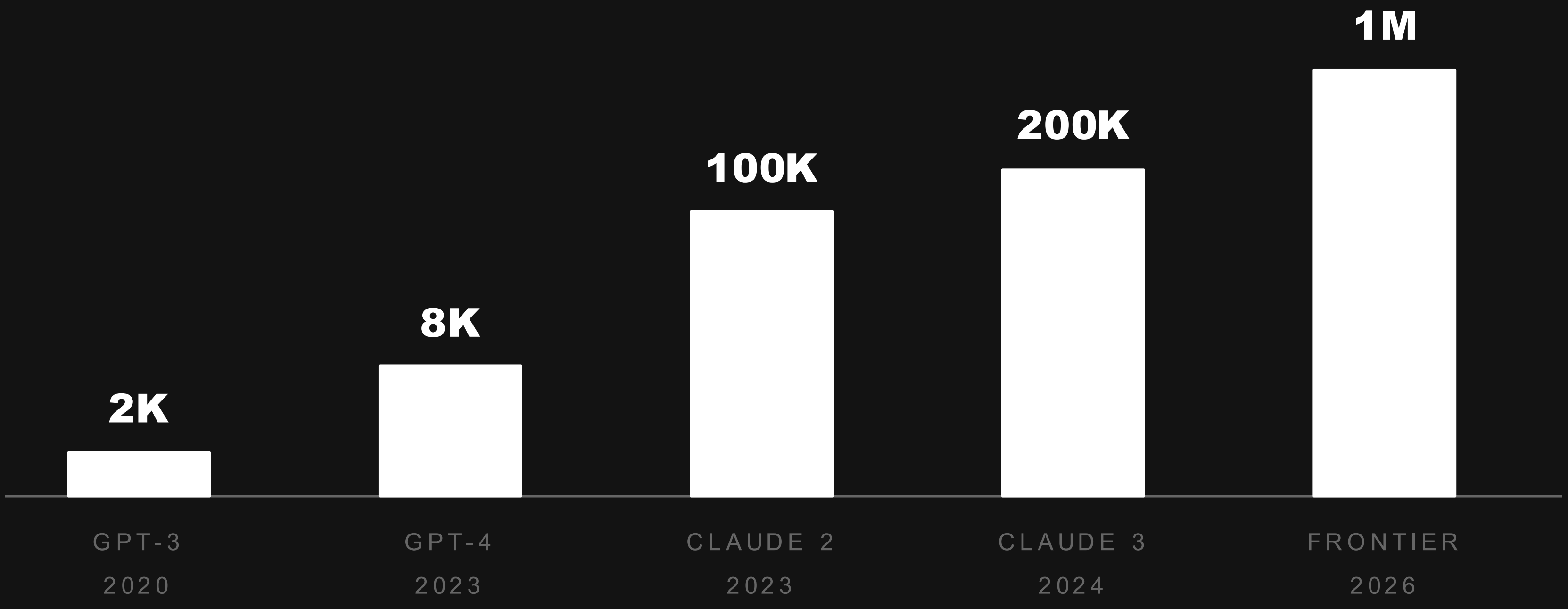
ALSO 2020

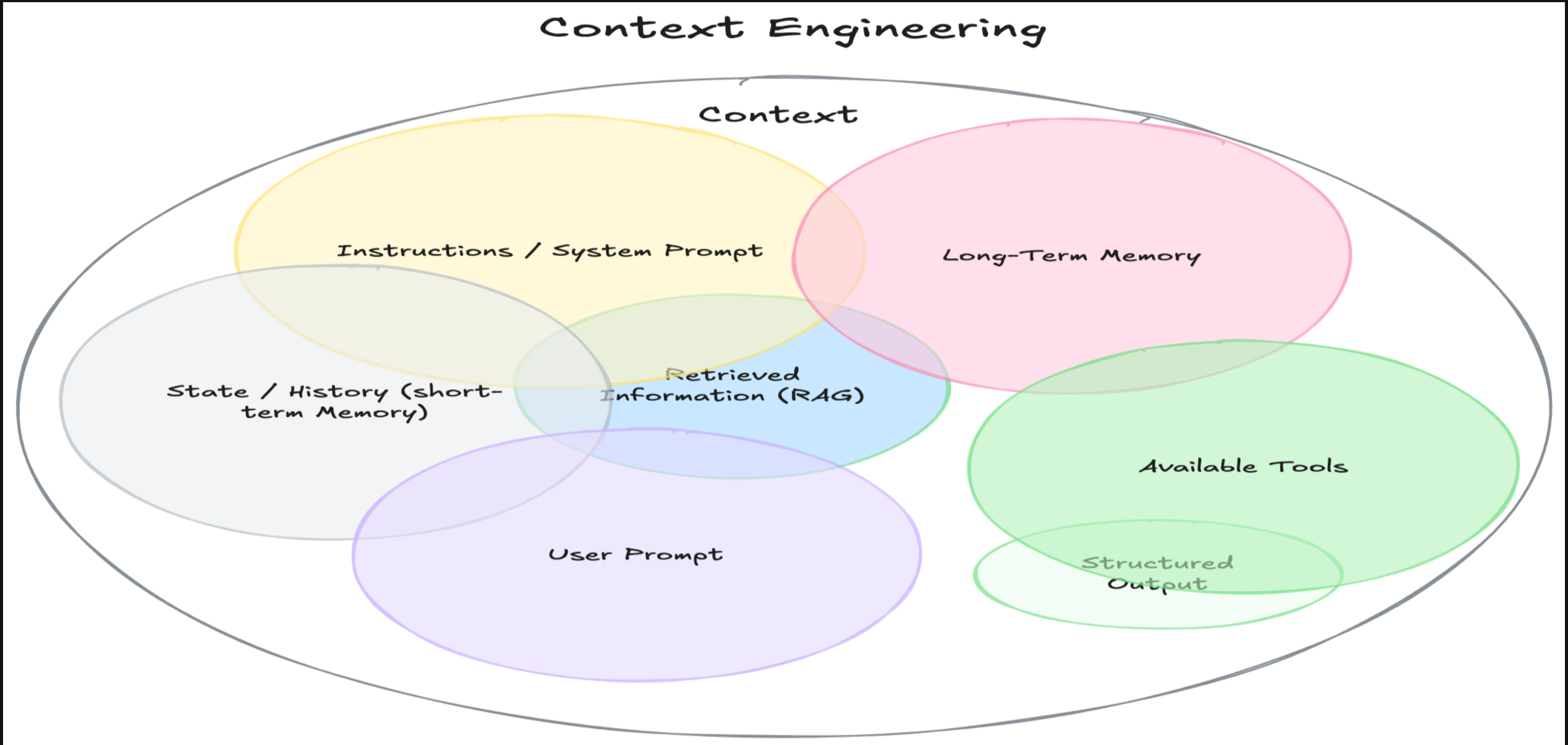
# RETRIEVE, THEN GENERATE.

*RAG — Lewis et al., FAIR. Fetch the facts at query time; let the model do the language.*

# The context window grew roughly 500× in six years.

*Tokens, log scale. 2,048 → ~1,000,000.*




Context Engineering
Context
Instructions / System Prompt
Long-Term Memory
State / History (short-term Memory)
Retrieved Information (RAG)
Available Tools
User Prompt
Structured Output

# 56.1%

Bury the answer in the middle of a long context and the model scores worse than with no documents at all.

*88.3% when given only the gold document. "Lost in the Middle," Liu et al., TACL 2024.*

so the job changed name

# curate. compress. order.

*Context engineering — "curating and maintaining the optimal set of tokens during inference." Named June 2025, defined by Anthropic that September.*

03

# HARNESS

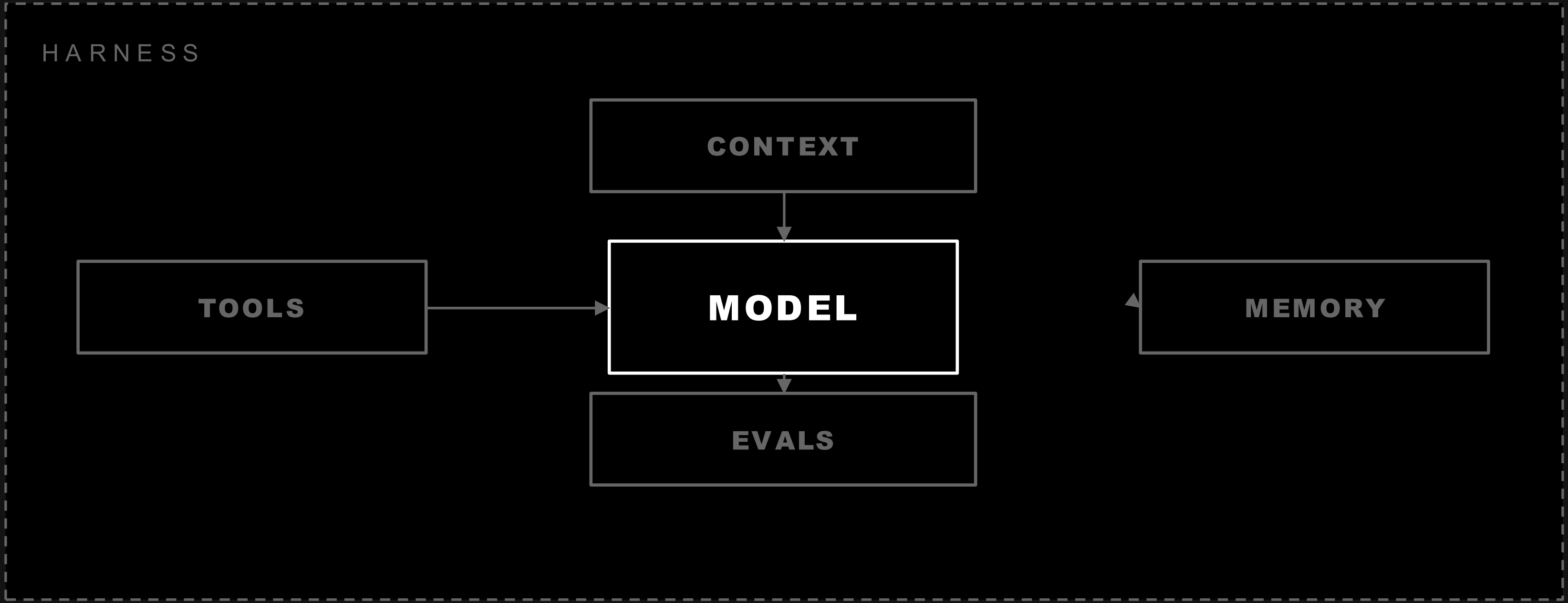
*what surrounds the model?*

an agent harness is

# the execution environment.

*It decides what context the model sees, what memory it keeps, what tools it can reach, and whether the output was any good. Everything outside the weights.*

# The model stopped being the product




*Curate the input, hold the state, expose the tools, judge the output.*

2 0 2 4

# ONE PROTOCOL FOR EVERY TOOL.

*MCP, released by Anthropic in November 2024. Donated to the Agentic AI Foundation in December 2025.*

04

# LOOP

## WHAT STARTED THE HYPE

Peter Steinberger
@steipete

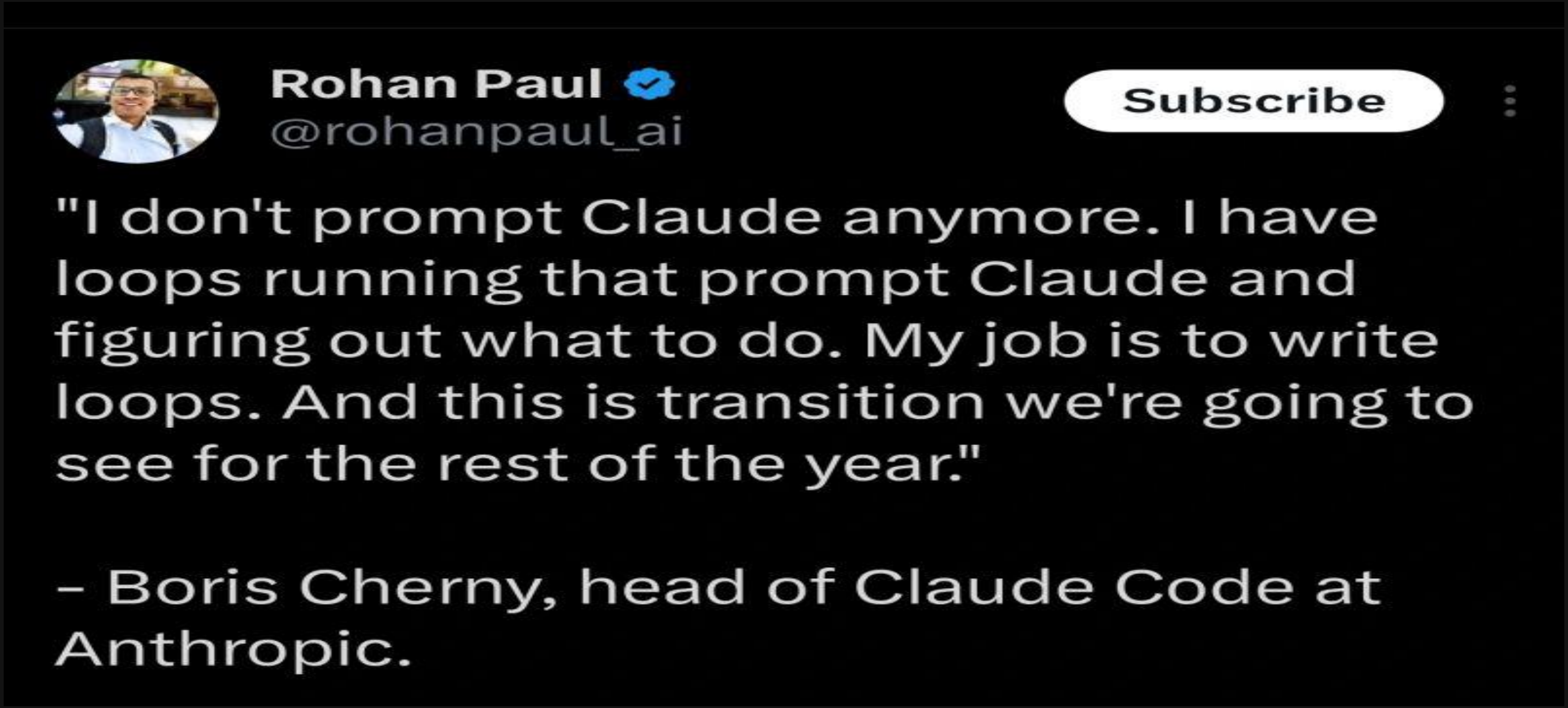
Here’s your monthly reminder that you shouldn’t be prompting coding agents anymore.

You should be designing loops that prompt your agents.

11:58 AM · Jun 7, 2026 · **1.2M** Views

803 906 9.6K 6.9K

## WHAT STARTED THE HYPE



THE DEFINITION

# A **repeating feedback loop** that allows an AI to **plan, act, evaluate, and improve** its own work until it reaches a desired outcome.

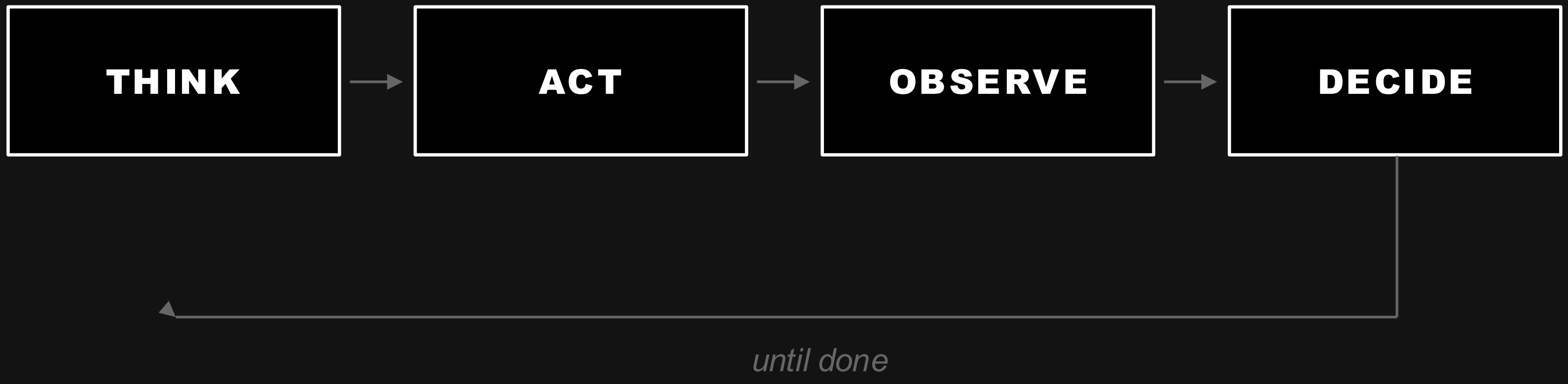
*Anthropic, "Building effective agents," December 2024. The whole idea, in five words.*

# Not one loop. Five shapes of loop.

| | | |
|---|---|---|
| 01 | **CORE AGENT** | Plan, act, evaluate, repeat until the task is done. |
| 02 | **VERIFICATION** | A second pass that checks the first: tests, critics, review. |
| 03 | **EVENT-DRIVEN** | No fixed start. It wakes on a webhook, a commit, a message. |
| 04 | **OPTIMIZATION** | Hold the goal fixed, vary the attempt, keep the best one. |
| 05 | **SELF-IMPROVEMENT** | The loop edits the agent itself: prompts, tools, memory. |

*Same primitive. Different trigger, different stopping rule.*

# One pass became many.




*The model no longer answers. It works.*

# 2×

Since 2024, the task length an agent can finish has doubled roughly every 88 days.

*At 50% reliability. METR, Time Horizon 1.1, January 2026.*

05

# GRAPH

*or is a graph just a loop that branches?*

# One word, two disciplines.

GRAPH AS DATA

## Entities and edges as the index.

*GraphRAG, Microsoft Research, open-sourced 2024.*

GRAPH AS CONTROL

## Nodes and edges as the program.

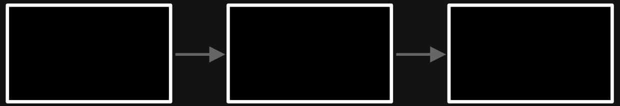
*LangGraph, LangChain. 1.0 GA in October 2025.*

*If you say "graph" without saying which one, nobody knows what you built.*

## WHAT STARTED THE HYPE

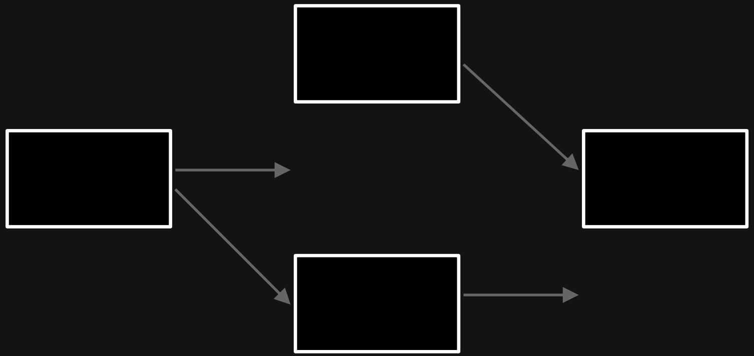
# Real systems stopped being a line.

A LOOP



one thread, one context

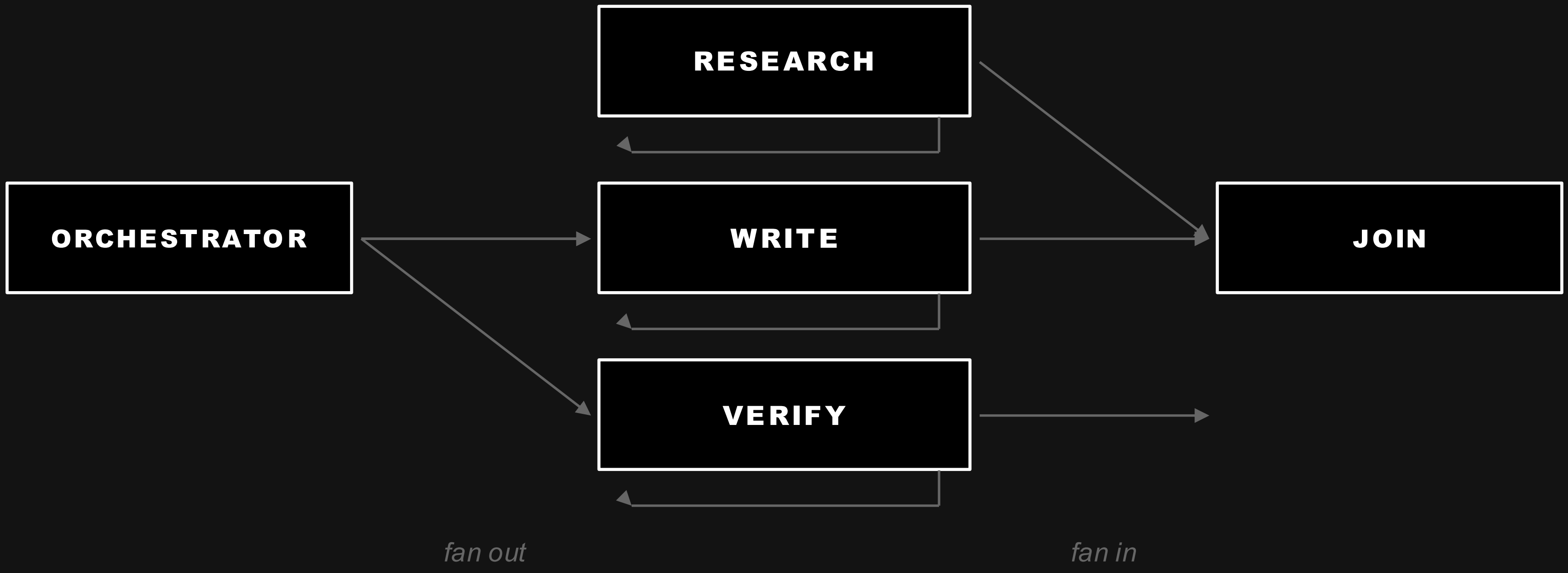
A GRAPH



branches, subagents, joins

*The practice arrived years before the label did.*

# The nodes aren't steps. They're loops.



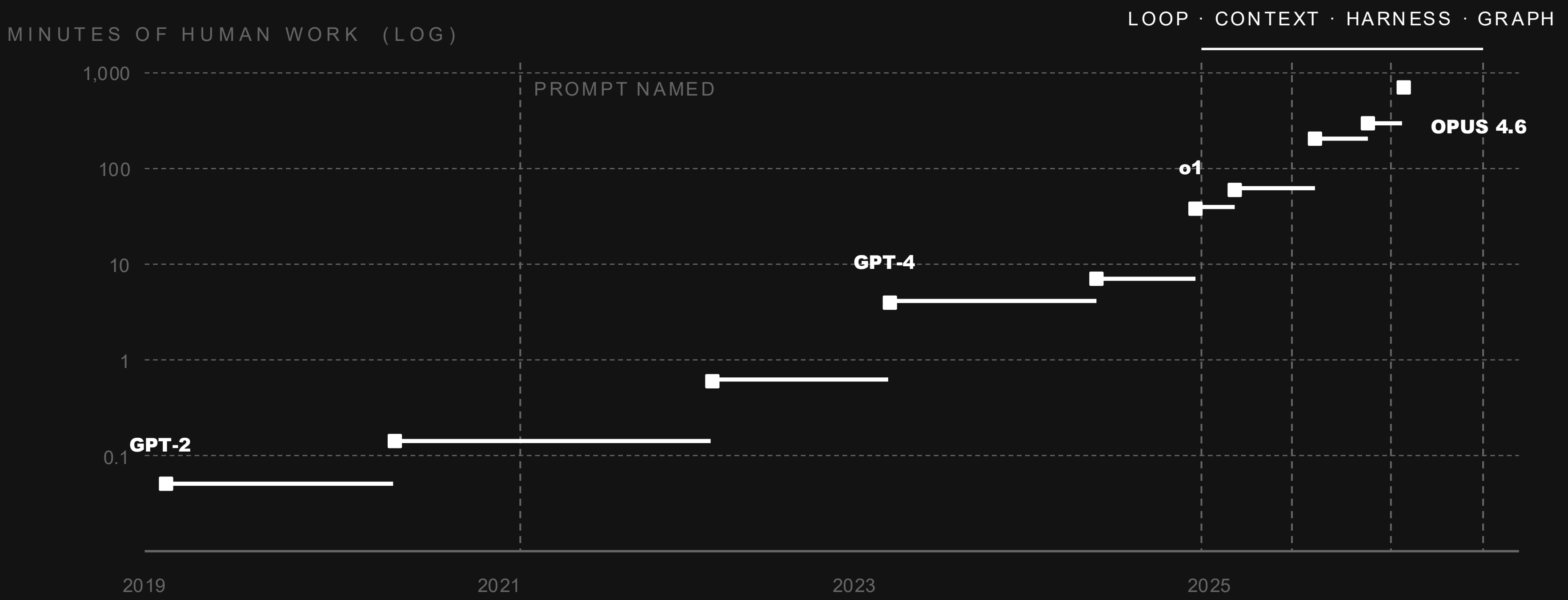

*Every node keeps running until it's done, then reports one answer upward.*

and the reason isn't control flow

# one window becomes four.

*Each loop works in its own context and hands back a summary. The orchestrator never sees the mess — which is a context problem solved with a shape.*

# The longer a model could work alone, the more we had to build around it.




3 seconds of work in 2019. Twelve hours in 2026 — and every layer arrived after the jump, not before.

*METR 50% time horizon, TH1.1, data to May 2026. METR calls its own measurements above 16 hr unreliable.*

WORKED EXAMPLE

# An agent that reviews pull requests.

| | |
|---|---|
| **CONTEXT** | The diff, the failing test output, and the three files it touches. Not the repo. |
| **HARNESS** | Sandboxed checkout, read-only git, a test runner it can call, a rubric it's graded on. |
| **LOOP** | Read, run tests, form a finding, check it against the code, revise. Stop when nothing new. |
| **GRAPH** | Three reviewers fan out — correctness, security, style. One node joins and dedupes. |

*Take away any one layer and it stops working.*

# ?

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